

Objective

To implement and demonstrate `StringIndexOutOfBoundsException` in Java using try-catch by accessing an invalid character position in a string.

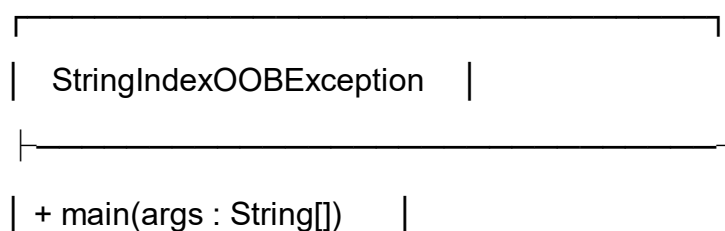
Steps for Execution

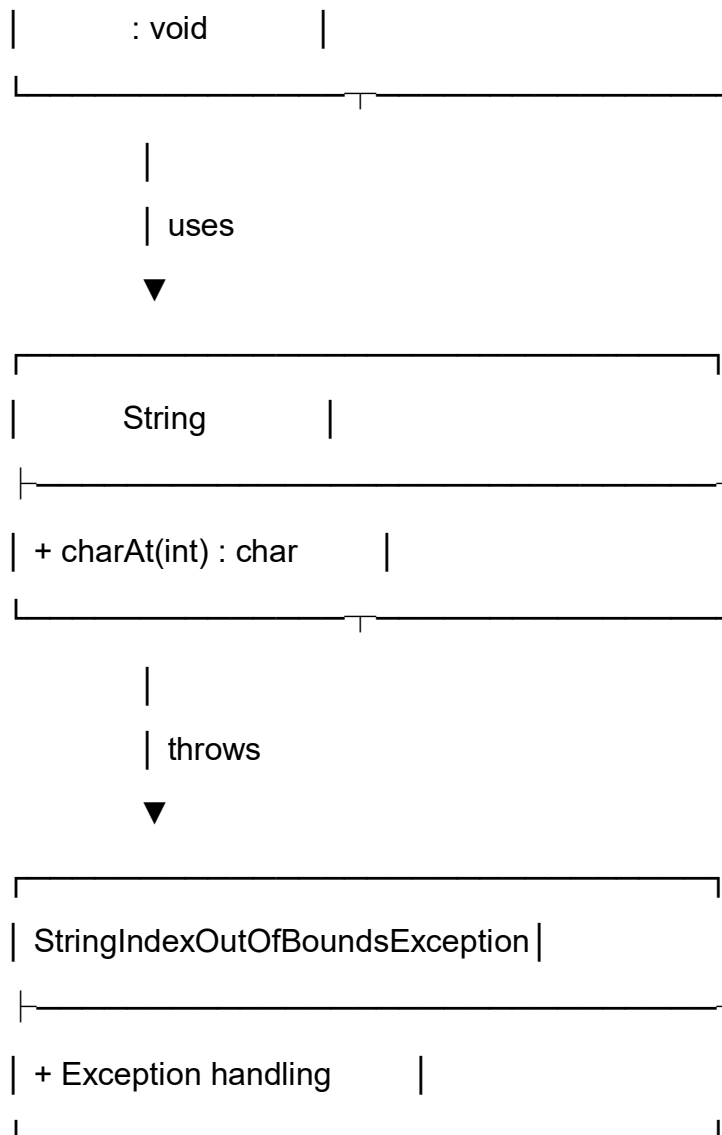
1. Create a Java project and package named `javacore`.
2. Create a file named `StringIndexOOBException.java` and enter the given code.
3. Declare a `String` variable with the value `"Java"`.
4. Access valid character positions using `charAt()`.
5. Try to access an invalid character position.
6. Handle the exception using catch.
7. Compile and run the program.
8. Observe the output.

Algorithm

1. Start the program.
2. Store `"Java"` in a `String` variable.
3. Display the `String` and valid characters.
4. Try to access character at index 10.
5. Catch `StringIndexOutOfBoundsException`.
6. Display the error message.
7. Display `"Program ended."`.
8. Stop the program.

UML Class Diagram





Relationship

- `StringIndexOutOfBoundsException` uses the `String` class and its `charAt()` method.
- The string "Java" has valid indexes from 0 to 3.
- Accessing index 10 causes a `StringIndexOutOfBoundsException`.
- The exception is handled using the catch block.

CODE:

```
package javacore;
```

```
public class StringIndexOutOfBoundsException {  
    public static void main(String[] args) {
```

```
String word = "Java";

try {
    System.out.println("Word: " + word);
    System.out.println("First character: " + word.charAt(0));
    System.out.println("Last character: " + word.charAt(3));

    System.out.println("Character: " + word.charAt(10));
}
catch (StringIndexOutOfBoundsException e) {
    System.out.println("Error: Character position is invalid.");
}

System.out.println("Program ended.");
}
```

Output

Word: Java

First character: J

Last character: a

Error: Character position is invalid.

Program ended.

Result

The program successfully demonstrates `StringIndexOutOfBoundsException` handling in Java. Since the string "Java" contains only four characters with indexes 0 to 3, accessing index 10 causes an exception, which is handled using the catch block.